



1. If $a = 1 + \log_x yz$, $b = 1 + \log_y xz$, $c = 1 + \log_z xy$, then $ab + bc + ca$ is equal to

NIMCET 2007

- (a) 0 (b) $2abc$
(c) abc (d) $a^2 + b^2 + c^2$
2. If $0 < y < 2^{1/3}$ and $x(y^3 - 1) = 1$, then $\frac{2}{x} + \frac{2}{3x^3} + \frac{2}{5x^5} + \dots =$

NIMCET 2007

- (a) $\log \left[\frac{y^3}{2-y^3} \right]$ (b) $\log \left[\frac{y^3}{1-y^3} \right]$
(c) $\log \left[\frac{2y^3}{1-y^3} \right]$ (d) $\log \left[\frac{y^3}{1-2y^3} \right]$
3. The value of $y = 0.36 \log_{0.25} \left(\frac{1}{3} + \frac{1}{3^2} + \dots \right)$ is

NIMCET 2008

- (a) 0.9 (b) 0.18 (c) 0.6 (d) 0.25
4. For $a > 0$, $a \neq 1$, the number of values of x satisfying the equation $2 \log_x(a) + \log_{ax}(a) + 3 \log_{a^2x}(a) = 0$

NIMCET 2008

- (a) 2 (b) 3 (c) 4 (d) 5
5. If $x < -1$ and $2^{|x+1|} - 2^x = |2^x - 1| + 1$, then the value of x is

NIMCET 2009

- (a) -2 (b) 2 (c) 0 (d) 1
6. The number of distinct integral values of 'a' satisfying the equation $2^{2a} - 3(2^{a+2}) + 2^5 = 0$ is

NIMCET 2009

- (a) 0 (b) 1 (c) 2 (d) 3
7. Solution set of the inequality $\log_3(x+2)(x+4) + \log_{1/3}(x+2) < \frac{1}{2} \log_{\sqrt{3}} 7$ is

NIMCET 2011

- (a) $(-2, -1)$ (b) $(-2, 3)$
(c) $(-1, 3)$ (d) $(3, \infty)$
8. If $x = \log_a bc$, $y = \log_b ca$ and $z = \log_c ab$, then $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$ is equal to

NIMCET 2012

- (a) abc (b) $\sqrt{ab} + \sqrt{bc} + \sqrt{ca}$
(c) 1 (d) $x + y + z$
9. If $2^a = 3^b = 6^{-c}$, then $ab + bc + ca$ is equal to

NIMCET 2012

- (a) 1 (b) 2 (c) 0 (d) None
10. If $x = \log_3 5$, $y = \log_{17} 25$, then which one of the following is correct?

- (a) $x > y$ (b) $x < y$ (c) $x \leq y$ (d) $x = y$
11. If $\log_x y = 100$ and $\log_2 x = 10$, then the value of y is

NIMCET 2013

- (a) 2^{10} (b) 2^{100} (c) 2^{1000} (d) 2^{10000}
12. If (x_0, y_0) is the solution of the following equations. $(2x)^{\ln 2} = 3y^{\ln x} = 2^{\ln y}$. Then, x_0 is

NIMCET 2014

- (a) $1/6$ (b) $1/3$ (c) $1/2$ (d) 6
13. If $a = \log_{12} 18$, $b = \log_{24} 54$, then $ab + 5(a - b)$ is

NIMCET 2014

- (a) 1 (b) 0 (c) 2 (d) $3/2$
14. If x , y and z are three consecutive positive integers, then $\log(1 + xz)$ is

NIMCET 2015

- (a) $\log y$ (b) $\log \frac{y}{2}$ (c) $\log(2y)$ (d) $2 \log(y)$
15. The solution set of equation $\log_x 2 \log_{2x} 2 = \log_{4x} 2$ is

NIMCET 2016

- (a) $\{2^{-\sqrt{2}}, 2^{\sqrt{2}}\}$ (b) $\left\{\frac{1}{2}, 2\right\}$
(c) $\left\{\frac{1}{4}, 2\right\}$ (d) $\left\{\frac{1}{4}, 2\right\}$

16. If $3^x = 4^{x-1}$, then x is equal to

NIMCET 2016

- (a) $\frac{2 - \log_3 2}{2 \log_3 2 - 1}$ (b) $\frac{2}{2 \log_3 2 - 1}$
(c) $\frac{2 - \log_3 2}{2 \log_3 2 + 1}$ (d) $\frac{2 \log_3 2}{2 \log_3 2 - 1}$

17. Sum of the roots of the equations $4^x - 3(2^{x+3}) + 128 = 0$ is

NIMCET 2016

- (a) 5 (b) 6 (c) 7 (d) 8

18. $6 + \log_{1/4} \frac{1}{\sqrt{2}} \left[\sqrt{1 - \frac{1}{\sqrt{2}}} \sqrt{1 - \frac{1}{\sqrt{2}}} \sqrt{1 - \frac{1}{\sqrt{2}}} \dots \right]$ is equal to

NIMCET 2018

- (a) 6 (b) $13/2$ (c) 4 (d) $25/4$

19. Solution set of the inequality $\log_3(x+2)(x+4) + \log_{1/3}(x+2) < \frac{1}{2} \log_{\sqrt{3}} 7$ is

NIMCET 2019

- (a) $(-2, -1)$ (b) $(-2, 3)$
(c) $(-1, 3)$ (d) $(3, \infty)$

20. The value of $3^{3 - \log_3 5}$ is

NIMCET 2022

- (a) $\frac{5}{27}$ (b) $\frac{27}{5}$ (c) $\frac{9}{5}$ (d) $\frac{5}{9}$

21. Which of the following number is the coefficient of x^{100} in the expansion of $\log_e \left(\frac{1+x}{1+x^2} \right)$, $|x| < 1$?

NIMCET 2023

- (a) 0.01 (b) 0.02 (c) -0.03 (d) -0.01

22. Let x be a positive real number such that



$$x^{(8 \log_5(x)-24)} = 5^{-4}.$$

Then the product of all possible values of x is

NIMCET 2025

- (a) 125 (b) 625
(c) 25 (d) 5

23. If $8^{x-1} = \left(\frac{1}{4}\right)^x$, then the value of $\frac{1}{\log_{x+1} 4 - \log_{x+1} 5} + \frac{1}{\log_{1-x} 4 - \log_{1-x} 5}$ is

NIMCET 2025

- (a) $\frac{5}{4}$ (b) 1 (c) 2 (d) $\frac{4}{5}$

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
c	a	b	a	a	c	b	c	c	a
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
c	c	a	d	a	d	c	b	b	b
21.	22.	23.							
a	a	c							

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	02	NIMCET 2016	03
NIMCET 2008	02	NIMCET 2017	00
NIMCET 2009	02	NIMCET 2018	01
NIMCET 2010	00	NIMCET 2019	01
NIMCET 2011	01	NIMCET 2020	00
NIMCET 2012	02	NIMCET 2021	00
NIMCET 2013	01	NIMCET 2022	01
NIMCET 2014	02	NIMCET 2023	01
NIMCET 2015	01	NIMCET 2024	00
		NIMCET 2025	02

SOLUTION

1. (c) $a = 1 + \log_x yz$

$b = 1 + \log_y xz$

$c = 1 + \log_z xy$

$a = \log_x x + \log_x yz = \log_x xyz$

$b = \log_y xyz$

$c = \log_z xyz$

$ab + bc + ca$

$(\log_x xyz)(\log_y xyz) + \log_y xyz \log_z xyz + \log_z xyz$

$\log_x xyz$

$$\frac{(\log xyz)^2}{\log x \cdot \log y} + \frac{(\log xyz)^2}{\log y \cdot \log z} + \frac{(\log xyz)^2}{\log z \cdot \log x}$$

$$(\log xyz)^2 \left\{ \frac{1}{\log x \cdot \log y} + \frac{1}{\log y \cdot \log z} + \frac{1}{\log z \cdot \log x} \right\}$$

$$(\log xyz)^2 \left[\frac{\log x + \log y + \log z}{\log x \cdot \log y \cdot \log z} \right]$$

$$\frac{(\log xyz)^3}{\log x \log y \log z} = \log_x xyz \cdot \log_y xyz \cdot \log_z xyz = abc$$

2. (a) $x(y^3 - 1) = 1$ Value of $\frac{2}{x} + \frac{2}{3x^3} + \frac{2}{5x^5} + \dots \infty$ we know that

$$\log \left(\frac{1+x}{1-x} \right) = 2 \left[x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \infty \right]$$

$$\ln \left[\frac{1+1/x}{1-1/2} \right] = \beta \left[\frac{1}{x} + \frac{1}{x^3} + \frac{1}{x^5} + \dots \infty \right]$$

$$= \ln \left[\frac{x+1}{x-1} \right] \because x = \frac{1}{y^3-1}$$

$$\ln \left[\frac{\frac{1}{y^3-1} + 1}{\frac{1}{y^3-1} - 1} \right] = \ln \left[\frac{y^3}{2-y^3} \right]$$

3. (b) $y = 0.36 \log_{\frac{25}{100}} \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots \right)$

$$y = 0.36 \log_{\frac{1}{4}} \left(\frac{\frac{1}{3}}{1 - \frac{1}{3}} \right) = 0.36 \log_{2^{-2}} (2^{-1})$$

$$y = 0.36 \log_{2^{-1}} 2^{-1} = \frac{0.36}{2} \log_2 2 = 0.4 \log_2 2$$

$$y = 0.18$$

4. (a) $a > 0, a \neq 1, x > 0, x \neq 1$

$$\Rightarrow \frac{2}{\log_a x} + \frac{1}{\log_a x} + \frac{3}{\log_a a^2 + \log_a x} = 0$$

$$\Rightarrow \frac{2}{\log_a x} + \frac{1}{\log_a a + \log_a x} + \frac{3}{\log_a a^2 + \log_a x} = 0$$

$$\Rightarrow \frac{2}{\log_a x} + \frac{1}{1 + \log_a x} + \frac{3}{2 + \log_a x} = 0$$

Let $\log_a x = t$

$$\Rightarrow \frac{2}{t} + \frac{1}{1+t} + \frac{3}{2+t} = 0$$

$$\Rightarrow \frac{2(1+t)(2+t) + t(2+t) + 3(1+t)t}{t(1+t)(2+t)} = 0$$

$$\Rightarrow 6t^2 + 11t + 4 = 0 \Rightarrow t = -\frac{1}{2}, -\frac{4}{3}$$

So, it has 2 solutions.

5. (a) $|x+1| = \begin{cases} x+1 & x+1 \geq 0 \Rightarrow x \geq -1 \\ -(x+1) & x+1 < 0 \Rightarrow x < -1 \end{cases}$



$$|2^x - 1| = \begin{cases} 2^x - 1 & 2^x \geq 1 \Rightarrow x \geq 0 \\ -(2^x - 1) & 2^x < 1 \Rightarrow x < 0 \end{cases}$$

$$2^{|x+1|} - 2^x = |2^x - 1| + 1 \quad [\text{As } x < -1]$$

$$\Rightarrow 2^{-x-1} - 2^x = -2^x + 1 + 1 \Rightarrow 2^{-x-1} = 2$$

On equating

$$\Rightarrow -x - 1 = 1 \Rightarrow x = -2$$

6. (c) $2^{2a} - 3(2^a \cdot 2^2) + 2^5 = 0$

$$\text{Let } (2^a = t > 0) \Rightarrow t^2 - 3(4)t + 32 = 0$$

$$\Rightarrow t^2 - 12t + 32 = 0 \Rightarrow (t - 4)(t - 8) = 0$$

$$\Rightarrow t = 4, 8 \Rightarrow 2^a = 2^2, 2^3$$

$$\Rightarrow a = 2, 3 \Rightarrow \text{two values}$$

7. (b) $\begin{cases} (x+2)(x+4) > 0 \\ (x+2) > 0 \end{cases}$

hence, $x > -2$

$$\log_3(x+2)(x+4) + \log_{3^{-1}}(x+2) < \frac{1}{2} \log_{3^{1/2}} 7$$

$$\Rightarrow \log_3(x+2)(x+4) - \log_3(x+2) < \log_3 7$$

$$\Rightarrow \log_3(x+4) < \log_3 7 \Rightarrow x+4 < 7 \Rightarrow x < 3$$

Therefore, the solution set of inequality is $(-2, 3)$.

8. (c) $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$

$$= \frac{1}{1 + \log_a bc} + \frac{1}{1 + \log_b ca} + \frac{1}{1 + \log_c ab}$$

$$= \frac{1}{\log_a a + \log_a bc} + \frac{1}{\log_b b + \log_b ca} + \frac{1}{\log_c c + \log_c ab}$$

$$= \frac{1}{\log_a abc} + \frac{1}{\log_b abc} + \frac{1}{\log_c abc}$$

$$= \log_{abc} a + \log_{abc} b + \log_{abc} c = \log_{abc} abc = 1$$

9. (c) Let $2^a = 3^b = 6^{-c} = k$

$$\Rightarrow 2^a = k, 3^b = k, 6^{-c} = k$$

$$\Rightarrow 2 = k^{\frac{1}{a}}, 3 = k^{\frac{1}{b}}, 6 = k^{\frac{1}{c}} \Rightarrow 2 \times 3 = k^{\frac{1}{c}}$$

$$\Rightarrow k^{\frac{1}{a}} \times k^{\frac{1}{b}} = k^{\frac{1}{c}}$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} = \frac{1}{c} \Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0 \Rightarrow \frac{ab + bc + ca}{abc} = 0$$

$$\Rightarrow ab + bc + ca = 0$$

10. (a) $x = \frac{\log 5}{\log 3}, y = \frac{\log 25}{\log 17}$

$$y = \frac{2 \log 5}{\log 17} = \frac{2 \log 5}{2 \log \sqrt{17}} = \frac{\log 5}{\log \sqrt{17}}$$

$$\text{As } \sqrt{17} > 3 \Rightarrow \log \sqrt{17} > \log 3$$

$$\Rightarrow \frac{1}{\log 3} > \frac{1}{\log \sqrt{17}} \Rightarrow \frac{\log 5}{\log 3} > \frac{\log 5}{\log \sqrt{17}}$$

$$\Rightarrow x > y$$

11. (c) $\log_2 x = 10 \Rightarrow x = 2^{10}$

$$\log_x y = 100 \Rightarrow y = x^{100} \Rightarrow y = (2^{10})^{100} \Rightarrow y = 2^{1000}$$

12. (c) (x_0, y_0) is solution of these two equations

$$(2x)^{\ln 2} = (3y)^{\ln 3}$$

Taking both sides of base e-

$$\ln(2x)^{\ln 2} = \ln(3y)^{\ln 3}$$

$$\Rightarrow (\ln 2)(\ln 2 + \ln x) = (\ln 3)(\ln 3 + \ln y)$$

$$\Rightarrow (\ln 2)^2 + \ln 2 \ln x = (\ln 3)^2 + \ln 3 \ln y \quad (i)$$

$$3^{\ln x} = 2^{\ln y}$$

Taking log both sides of base e,

$$\ln x \ln 3 = \ln y \ln 2$$

$$\ln x = \frac{\ln y \ln 2}{\ln 3} \quad (ii)$$

Put this value in Eq. (i),

$$\ln 2 \left(\ln 2 + \frac{\ln y \ln 2}{\ln 3} \right) = \ln 3 (\ln 3 + \ln y)$$

$$\ln y \left[\frac{(\ln 2)^2}{\ln 3} - \ln 3 \right] = (\ln 3)^2 - (\ln 2)^2$$

$$\text{Hence, } \ln y = -(\ln 3) = \ln \frac{1}{3}$$

$$\text{Hence, } y = 1/3$$

Put this value in Eq. (ii)

$$\ln x = \frac{\ln y \ln 2}{\ln 3} = \frac{(-\ln 3) \ln 2}{\ln 3} = -\ln 2$$

$$\text{Hence, } x = 1/2.$$

13. (a) We have

$$a = \log_{12} 18 = \frac{\log_2 18}{\log_2 12} = \frac{\log_2 (2 \times 3^2)}{\log_2 (2^2 \times 3)} = \frac{\log_2 2 + 2 \log_2 3}{\log_2 2^2 + \log_2 3}$$

$$a = \frac{1 + 2 \log_2 3}{2 + \log_2 3}$$

$$b = \log_{24} 54 = \frac{\log_2 54}{\log_2 24} = \frac{\log_2 (2 \times 3^3)}{\log_2 (2^3 \times 3)} = \frac{1 + 3 \log_2 3}{3 + \log_2 3}$$

$$\text{Let } \log_2 3 = t$$

$$\text{Hence, } ab + 5(a - b)$$

$$= \left(\frac{1+2t}{2+t} \right) \left(\frac{1+3t}{3+t} \right) + 5 \left(\frac{1+2t}{2+t} - \frac{1+3t}{3+t} \right) = \frac{t^2 - 5t + 6}{(t+2)(t+3)} = 1$$

14. (d) $x = n - 1, n \in N, y = n$ and $z = n + 1$



$$\log(1+xz) = \log(1+(n-1)(n+1))$$

$$= \log(1+(n^2-1)) = \log n^2 = 2 \log n = 2 \log y$$

15. (a) $x > 0, x \neq 1, \frac{1}{2}, \frac{1}{4}$

$$\frac{\log_2 2 \log_2 2}{\log_2 x \log_2 2x} = \frac{\log_2 2}{\log_2 4x}$$

$$\Rightarrow \frac{1}{\log_2 x (1 + \log_2 x)} = \frac{1}{(2 + \log_2 x)}$$

$$\Rightarrow \frac{1}{t} \times \frac{1}{(1+t)} = \frac{1}{2+t}$$

$$\Rightarrow 2+t = t^2+t \Rightarrow t^2 = 2$$

$$\Rightarrow t = \pm\sqrt{2} \Rightarrow \log_2 x = \sqrt{2} \text{ and } \log_2 x = -\sqrt{2}$$

$$\Rightarrow x = 2^{\sqrt{2}}, 2^{-\sqrt{2}}$$

16. (d) Given, $3^x = 4^{x-1}$

Taking log both sides of base 3,

$$\log_3 3^x = \log_3 4^{x-1}$$

$$\Rightarrow x = (x-1)\log_3 4 \Rightarrow x = (x-1)\log_3 2^2$$

$$\Rightarrow x = 2(x-1)\log_3 2$$

$$\Rightarrow x = (2x-2)\log_3 2 \Rightarrow x = 2x \log_3 2 - 2 \log_3 2$$

$$\Rightarrow 2 \log_3 2 = 2x \log_3 2 - x$$

$$\Rightarrow 2 \log_3 2 = x(2 \log_3 2 - 1) \Rightarrow x = \frac{2 \log_3 2}{(2 \log_3 2 - 1)}$$

17. (c) $(2^x)^2 - 3(2^x \cdot 2^3) + 2^7 = 0$

$$\text{Let } 2^x = t > 0$$

$$\Rightarrow t^2 - 3 \cdot 2^3 t + 2^7 = 0 \Rightarrow t^2 - 24t + 128 = 0$$

$$\Rightarrow (t-16)(t-8) = 0 \Rightarrow t = 16, 8 \Rightarrow 2^x = 2^4, 2^3$$

$$\Rightarrow x = 4, 3 \Rightarrow \text{Sum of roots} = 4 + 3 = 7$$

18. (b) Let $x = \frac{1}{\sqrt{2}} \sqrt{1 - \frac{1}{\sqrt{2}} \sqrt{1 - \frac{1}{\sqrt{2}} \sqrt{1 - \frac{1}{\sqrt{2}}}}}$

$$\Rightarrow x = \frac{1}{\sqrt{2}}(1-x) \Rightarrow x^2 = \frac{1}{2}(1-x), \text{ We know } x > 0$$

$$\Rightarrow 2x^2 = 1-x \Rightarrow 2x^2 + x - 1 = 0$$

$$\Rightarrow 2x^2 + 2x - x - 1 = 0 \Rightarrow (2x-1)(x+1) = 0 \Rightarrow x = \frac{1}{2}, -1$$

$$\text{As } x > 0 \Rightarrow x = \frac{1}{2}$$

$$6 + \log_{2^{-1}} 2^{-1} = 6 + \frac{1}{\log_2 2} = \frac{6}{1} + \frac{1}{2} = \frac{13}{2}$$

19. (b) Refer to solution 5.

20. (b) $3^3 \times 3 - \log_3 5 = 3^3 \cdot 3^{\log_3 5^{-1}} = 3^3 \times 5^{-1} = \frac{27}{5}$

21. (a) $\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots - \frac{x^{100}}{100} + \dots$

$$\ln(1+x^2) = x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \frac{x^8}{4} + \dots - \frac{(x^2)^{50}}{50}$$

Coefficient of x^{100} in $\ln(1+x) - \ln(1+x^2)$

$$= -\frac{1}{100} + \frac{1}{50} = 0.01$$

22. (a)

Let $x > 0$. Put $t = \log_5 x \Rightarrow x = 5^t$

$$x^{8 \log_5 x - 24} = (5^t)^{8t-24} = 5^{t(8t-24)} = 5^{-4}$$

So

$$8t^2 - 24t = -4 \Rightarrow 8t^2 - 24t + 4 = 0$$

$$\Rightarrow 2t^2 - t + 1 = 0$$

Roots:

$$t = \frac{6 \pm \sqrt{36-8}}{4} = \frac{3 \pm \sqrt{7}}{2}$$

Hence $x = 5^t = 5^{(3 \pm \sqrt{7})/2}$

Product of all possible x :

$$x_1 x_2 = 5^{t_1+t_2} = 5^3 = 125$$

23. (c) Given: $8^{x-1} = \left(\frac{1}{4}\right)^x$ rewrite as $(2^3)^{x-1} = (2^2)^{-x}$

$$2^{3x-3} = 2^{-2x} \text{ equate exponents:}$$

$$3x - 3 = -2x \Rightarrow 5x = 3 \Rightarrow x = \frac{3}{5}$$

Evaluate $S = \frac{1}{\log_{x+1} 4 - \log_{x+1} 5} + \frac{1}{\log_{1-x} 4 - \log_{1-x} 5}$

Substitute $x = \frac{3}{5}$, so $x+1 = \frac{8}{5}$, $1-x = \frac{2}{5}$

Using $\log_b a = \frac{\log a}{\log b}$:

$$S = \frac{1}{\frac{\log 4 - \log 5}{\log \frac{8}{5}}} + \frac{1}{\frac{\log 4 - \log 5}{\log \frac{2}{5}}} = \frac{\log \frac{8}{5} + \log \frac{2}{5}}{\log 4 - \log 5} = \frac{\log \frac{16}{25}}{\log 4 - \log 5} =$$

$$\frac{4 \log 2 - 2 \log 5}{2 \log 2 - \log 5} = 2$$

Ans. 2